Representing Inequalities on a Number Line

AQA · KS4 Year 9–10

Answer each question in your workbook. Draw each number line carefully.

Method

Find the boundary number   Open   for </>, filled   for   /   Arrow towards the true values

Shade between for 'between' inequalities   Read back to check

Common mistakes

- Filled circle for < or > (should be open).
- Open circle for   or   (should be filled).
- Arrow pointing the wrong way.
- -2   x is the same as x   -2.

Worked example

Represent x   -2.

includes -2, so a filled circle; arrow points right.

Read-back: filled   at -2, arrow right = x   -2

Visual aid

Visual Example

Open circle   means the boundary value is NOT included (< or >). Filled circle means it IS included (or   ). The arrow points towards all the numbers that make the inequality true.

Questions

Answer each question in your workbook. Draw each number line carefully.

Q1 of 8 [2]

Show on a number line: a) $x > 1$ b) x 4

Q2 of 8 [2]

Show on a number line: a) x -3 b) $x < 0$

Q3 of 8 [2]

Write the inequality: a) open circle at 2, arrow right b) filled circle at -1, arrow left

Q4 of 8 [3]

Show on a number line: a) x 5 b) $x < -4$ c) -2 x

Q5 of 8 [3]

Solve, then show on a number line: a) $2x + 1 > 7$ b) $3x - 2$ 10 c) $5 - 2x$ 1

Q6 of 8 [3]

Write and show: a) the number n is at least -1 b) the temperature t (°C) is below 5

Q7 of 8 [3]

Filled circle at -3, open circle at 4, shaded between. a) Write the inequality. b) List the integers.

Q8 of 8 [5]

Machine safe when t   10 and t < 35. a) Show each on a number line. b) Combine into one inequality. c) Is 35 °C safe? Explain.